

Properties of Logarithms (3.3)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 3.322 # _____ Evaluate $\log_2 8$.	ANSWER: 4 # _____ Evaluate $\log_2 \frac{1}{8}$.
ANSWER: 0 # _____ Simplify $\log_6 2 + \log_6 3$.	ANSWER: 12 # _____ Condense to a single logarithm: $2 \log x + \log 3$.

ANSWER: 3 # _____ Evaluate $\log_5 1$.	ANSWER: -3 # _____ Simplify $5^{\log_5 12}$.
ANSWER: $\frac{1}{2}$ # _____ Simplify $\log_4 32 - \log_4 2$.	ANSWER: $\log(3x^2)$ # _____ (Calculator) Use change of base to evaluate $\log_2 10$ to the nearest thousandth.
ANSWER: 1 # _____ Evaluate $\log_5 \sqrt{5}$.	ANSWER: 2 # _____ Evaluate $\log_3 81$.